Nagur Shareef Shaik

Applied Scientist | Multimodal Medical AI & Agentic Systems

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Summary

Applied Scientist with 6.5+ years building medical AI that holds up on messy, incomplete patient data. My PhD research at Georgia State targets what breaks multi-modal models in practice: **missing modalities**, **imbalanced classes**, and predictions that ignore how a patient changes over time. That work has led to **29** peer-reviewed papers (AAAI, NeurIPS, ICASSP, ICIP), **1.4K+** citations, and an H-index of **14**. At UST, I lead an **18-person team** that ships it: agentic and RAG systems across healthcare, retail, and finance, from claims and coding to AI-assisted pathology and legacy modernization.

Education

Georgia State University — Atlanta, GA
Doctor of Philosophy in Computer Science | GPA: 4.23/4.30 — *Aug 2023 – Dec 2026*

Vignan's Foundation for Science, Technology & Research University — Guntur, AP
Bachelor of Technology (Honours) in Computer Science & Engineering | GPA: 3.92/4.0 — *Jun 2016 – May 2020*

Experience

Georgia State University — Aug 2025 – Present
Graduate Research Assistant | TReNDS Center — *Atlanta, GA*

- Lead my doctoral research on **multi-modal medical AI that holds up when data is incomplete**: modality-aware perception, latent disentanglement under missing modalities, and longitudinal clinical reasoning that keeps track of uncertainty.
- Developed a **probabilistic multi-modal framework** (HSIC-constrained latent factorization plus a Self-Critiquing Mixture-of-Experts) that set **SOTA** across **7 systemic & ocular conditions**.

UST Global Inc. — July 2024 – Present
Machine Learning Engineer II | AI-CoE — *(Remote) Aliso Viejo, CA*

- Lead an **18-person AI team** in the AI-CoE building **UST Alpha AI**, UST's one-stop shop for AI workloads, and its platform-modernization tools. Led development of **CodeCrafter**; **designed and built Orbit** myself.
- Run delivery on an **AI-DLC** approach: agents migrate legacy **.NET 4→8** and **COBOL** monoliths to **React/Java Spring Boot**, generating **NX monorepos** from specs that land **95%** functional, with delivery **70%** faster at **60%** of the cost. Now reimagining a **NextGen Reinsurance Platform** with **SpecKit**; the MVP already works.
- Built **PathForce**, an AI-assisted **anatomic-pathology LIS** (rules-first eligibility checks, effort-weighted case distribution, **HL7/FHIR/X12**, **Part 11-grade** audit trail), plus **RiskIQ** for **ICD-10 coding & risk adjustment** and **PolicyIQ**, which turns **CMS/payer policy** into SME-approved claim edits in minutes, not weeks.
- Modernized a payer's **claims adjudication** by re-engineering its extracts and reporting, which improved **payment accuracy** and audit-readiness and cut manual reconciliation and reprocessing. Architected and led **Call Analytics Agents** that raise **First-Call Resolution**, reduce repeat contacts, and flag quality & compliance issues.
- Architected **BSA DataMap** (agentic profiling into **BigQuery**, saving **2 FTEs/$250K**) and an **ETL Test Agent** validated with **LLM-as-a-Judge**. Delivered **RegView**, natural-language search over regulatory documents, and a **RAG-powered RFI/RFP engine** that cut proposal time by **60%**.

Georgia State University — Sep 2023 – May 2025
Graduate Research Assistant | TReNDS Center — *Atlanta, GA*

- Customized **multi-modal LLMs** in **PyTorch** to generate reports from medical images and text: **13.4% higher** BLEU-4 than VisionGPT, with inference down to **1.6 sec** per image.
- Built **TensorFlow/Keras** image classifiers for retinopathy, schizophrenia, breast cancer, and colon cancer, cutting false negatives by **5%–7%**.

Carelon Global Solutions — Sep 2022 – Aug 2023
Software Engineer III | Elevance Health — *Hyderabad, TS*

- Built **REST APIs** for the **CO**mpensation **IN**centive **S**ystem, a **microservices** application that validates, computes, and speeds up incentive payments, cutting processing time by **10%**. Wrote **Python/SQL** clean-up scripts that stopped over **$500K** in overpayments.
- Fixed critical production issues in the commissions calculation flow (agent advance computation), saving **$1.5 million** in historical overpayments; earned the **Go Above – Impact Award**.

Tata Consultancy Services Aug 2020 – Sep 2022
Systems Engineer | Analytics & Insights Business Unit *Hyderabad, TS*

- Designed a log-based Bug Root Cause Prediction System with an **Attention LSTM** in **Azure ML Studio**, cutting debugging effort and saving **3 full-time equivalents** a year.
- Set up an **Azure DevOps** model-deployment pipeline that took deployment from **2 to 1.2 hours** and raised system availability by **25%**; named team **Star Performer**.

Doctoral Dissertation

From Perception to Clinical Reasoning: Structured & Disentangled Latent Modeling for Robust Multi-Modal Medical AI | Advisor: **Dong Hye Ye** Aug 2023 – Dec 2026

- **Thesis:** reliable clinical reasoning from incomplete, heterogeneous, longitudinal data needs structured representations, uncertainty carried through time, and evaluation that separates visual evidence from concept priors. Validated across **7 clinical domains**, **13 benchmarks**, and **8 novel architectures** (proposal defended Apr 2026).

Structured Multi-Modal Perception *Modality-aware attention & fusion*

- Designed **modality-aware attention** that respects how clinical data is structured, beating generic attention by **2–7%** across retinal, endoscopic, and neuroimaging tasks.
- Built **asymmetric vision–text–genomics fusion** that reaches **SOTA** on report generation and classification (**0.231** BLEU-4; **86.05%** accuracy).

Robust Latent Disentanglement *Missing modalities, imbalance, collapse*

- Introduced **probabilistic latent disentanglement** (VLVAE, WAE, Mixture-of-Experts) so models degrade gracefully under **missing modalities**, **class imbalance**, and **representation collapse**.

Uncertainty-Aware Longitudinal Reasoning *Distributional temporal modeling*

- Developed **uncertainty-aware longitudinal reasoning** that treats disease progression as **distributional change on statistical manifolds** (Fisher–Rao), so diagnostic ambiguity survives and claims stay tied to image evidence; **SOTA +7.4% CE-F1** on Longitudinal MIMIC-CXR. *(Dist-RRG: ICLR 2027, under review; STRIDE: AAAI 2027, under review; DELTA-E: CVPR 2027, in preparation)*

Master's Thesis

Attentive Multi-modal Learning for Medical Image Analysis | Advisor: **Dong Hye Ye** Aug 2023 – May 2025

- Advanced **multi-modal LLMs** and **attention networks** that combine medical images, clinical text, and genetics to improve schizophrenia diagnosis and automate report generation.

Skills

Languages: Python, Java, SQL
Technologies: LangGraph, LangChain, Flask, FastAPI, Streamlit, Tableau, Spring Boot, REST APIs, Microservices
Agentic Tools: Claude Code, Codex, Cursor, Windsurf, GitHub Copilot
Libraries: PyTorch, torchtune, TensorFlow, Keras, Scikit-Learn, OpenCV, NLTK, NumPy, Pandas, Matplotlib
Cloud & MLOps: MLflow, AWS (EC2, Lambda, SageMaker, Bedrock AgentCore), Azure (AI Services, DevOps), GCP (Agentspace, Gemini Enterprise), CI/CD, Git, Docker
Research Interests: Deep Learning, Multi-modal Learning, Attention Networks, Large Language Models, Agentic AI
Certifications: Azure AI Fundamentals, Deep Learning Specialization, Python for Everybody

Projects

PathPro | GitHub | Demo | *Python, FastAPI, React, LightGBM, MapLibre* Sep 2026

- Built an explainable pedestrian traffic-risk forecaster for Atlanta (**Poisson GLM + LightGBM**, **Empirical Bayes**, weather-aware scoring) with risk-aware routing and citywide maps. Reaches **0.869 ROC-AUC** on the **2023 temporal holdout**, with exact risk-factor decomposition and validated AI explanations.

InTune | GitHub | Demo | *Next.js, TypeScript, LangGraph.js, Gemini, LiveKit* Sep 2026

- Built a communication-support community for **text, audio, and video**. **LangGraph.js** and **Gemini** handle clarification, meaning-preserving rewrites, and private reading aids; exact text-and-audience approval, critical-slot checks, and idempotent publishing keep it safe, and **LiveKit** powers calls with private AI interpretation.

InsightRx | GitHub | Demo | *DINOv2-L, LoRA, FastAPI, ONNX* Sep 2026

- Built a retinal screening and HCP collaboration workspace on a **DINOv2-L/LoRA ensemble** (**0.980 patient-level AUROC** for referable DR on **247 held-out mBRSET patients**), with quality-gated two-eye assessment, attention maps, clinician-signed referrals, and traceable links to guideline therapy, protein targets, and drug evidence.

AURA | GitHub | *Python, FastAPI, React/TypeScript, Vision Transformers, Gemini, ChromaDB* Feb 2026
- Built a multi-agent healthcare ecosystem where a master **intent router** hands off to specialist agents (diagnosis, CBT wellbeing, triage, nutrition, visualization), including **IRIS**, a **Vision Transformer** that predicts systemic risk from a mobile retinal image, and **COMPASS**, an **Agentic RAG** insurance guide. Benchmarked on mBRSET, MIMIC-CXR, DeepEyeNet, and ROCO.

PRISM | GitHub | *Python, FastAPI, React, LangGraph, LLMs, Vector DB* Oct 2025
- Built an **AI-powered data-intelligence platform** that combines **NL2SQL**, automated ETL, one-click ML pipelines, no-code **RAG**, and auditable agentic workflows.

NL2Dash | GitHub | *Python, NL2SQL, LLMs, FastAPI, React* Sep 2025
- Built a **self-serve analytics** tool that turns plain-English questions into governed SQL (**NL2SQL**) and person-specific dashboards, so ad-hoc reporting no longer waits on the BI team.

AgentKit | GitHub | *Python, LangGraph, Agentic Frameworks, Observability* Aug 2025
- Built an **agent-orchestration library** that adds a production **harness**, pluggable **memory**, **observability**, and **evaluation** to open-source agentic frameworks, shortening time-to-production across teams.

InsuCompass | GitHub | Demo | *Python, LangGraph, Streamlit, Groq, Gemini, ChromaDB* May 2025
- Built an **Agentic RAG** health-insurance advisor whose self-learning **Search Agent** retrieves, verifies, and ingests new knowledge (CMS.gov, VA.gov); orchestrated in **LangGraph** on **Llama 3** and **Gemini**.

ScholarPulse | GitHub | Demo | *Python, LangChain, Streamlit, ChromaDB, SentenceTransformers* Apr 2025
- Built an **Advanced RAG** research assistant (**LangChain**, **Llama-4 Maverick**, **Qwen-2.5**) for paper Q&A, summarization, and code generation, cutting research comprehension time by **50%**.

Retinal Health Diagnostics | GitHub | Demo | *Python, TensorFlow, FastAPI, Streamlit* May 2024
- Developed a novel attention mechanism that uses global context to learn localized, lesion-specific features: **97.5%** (cataract), **85.6%** (diabetic retinopathy), and **94.6%** (macular edema) accuracy.

Research & Publications

- Published **29** peer-reviewed papers (**9** conference; **20** journal and workshop; **1.4K+ citations**, **H-index 14**, Google Scholar), with **oral presentations** at **AAAI 2026**, **EMBC 2025 & 2026**, **MLSP 2026**, **ICIP 2024**, and **ISBI 2024**, plus papers at **ICASSP 2025**, **ISBI 2025**, and **NeurIPS 2025** (GenAI4Health).
- Collaborated with **Dr. Tatsunori Hashimoto** (**Stanford AI Lab**) on **AxisSQL** (**inference-time scaling**), a **Text-to-SQL** reasoning framework ranked **#12 on the BIRD leaderboard** at submission (Leaderboard); **submitted to EACL (Industry Track)**.
- Collaborated with **Dr. Ehsan Adeli** (Stanford; through UST) on work at the **NeurIPS 2025 GenAI4Health Workshop**.
- Reviewer for **25+ journals** and major conferences (AAAI, MICCAI, NeurIPS, EACL, ISBI), with **60+ papers** reviewed across AI/ML, biomedical imaging, and computer vision.

Accomplishments

- **Graduate Research Award 2026**, Georgia State University; winner, **D3CODE Hackathon 2025** (US Region & Global), UST Global Inc., ahead of teams from the UK, Mexico, Malaysia, and India.
- **STELLER**, **NOVA Star** and **Super Star** Awards, UST Global Inc.; **Go Above – Impact Award**, Elevance Health (Carelon Global Solutions); **Star Performer Award**, Tata Consultancy Services.
- **NeurIPS 2025 GenAI4Health** and **IEEE SPS (ICIP 2024)** Travel Grants; **100% graduate scholarship**, Georgia State University; **Chairman's Gold Medal**, **Best Outgoing Student of CSE**, and **4-year Academic Excellence Awards**, VFSTR University.